

Comparative Analysis of SVM and Decision Trees for Banking Data

We summarize and compare the performance of two distinct modeling approaches: Support Vector Machines (SVM) and Decision Trees for predicting client bank deposit subscriptions. The primary challenge in this analysis is the dataset's significant class imbalance, with the 'yes' (subscribe) class representing a small minority.

Importantly, this analysis compares models that were both explicitly configured to handle this imbalance using the **class_weight='balanced'** parameter to ensure a fair and direct comparison.

The analysis shows that while the Linear SVM model has a marginal edge in pure recall, a pruned Decision Tree (specifically with **min_samples_leaf=35** parameter) delivers statistically identical performance across key metrics (F1-Score, AUC). The DT offers slightly better precision and a significant advantage in model interpretability for this use case.

Support Vector Machine (SVM) Model Analysis

The SVM analysis predicts client subscriptions using two distinct kernels: Linear and Radial Basis Function (RBF).

Handling Class Imbalance

A core component of the SVM strategy was to directly address the dataset's skew. This was accomplished by implementing the **class_weight='balanced'** parameter in both the Linear and RBF models. This technique modifies the loss function to give a higher penalty to misclassifications of the minority 'yes' class; this forces the model to prioritize their identification.

Additionally, in the initial data preparation step, the train-test split was stratified to ensure that both the training and test sets maintained the same, small percentage of the minority 'yes' class as the original dataset. This prevents the model from being trained or tested on an unrepresentative sample, ensuring the performance metrics are reliable.

Hyperparameter Tuning and Performance

A significant performance difference was observed in training time. The non-linear RBF kernel was computationally intensive. Due to limited, but standard, desktop compute on multi-core Apple Silicon, SVM with Grid Search training never completed on the data set of this size, using the number of iterations required for the Grid Search tuning. To address performance and training time limitations, we used a subsampled RandomizedSearchCV to efficiently tune the C and gamma hyperparameters. This RandomizedCV search was performed on a stratified 10,000-record subsample to find optimal

parameters without incurring the prohibitive cost of a full grid search. This improved performance and reduced training time significantly (for the purposes of this report).

SVM Results and Recommendation

After tuning, both the Linear SVM and the RBF SVM produced nearly identical results for the minority 'yes' class.

Model Performance Comparison:		
	Linear SVM	RBF SVM
Metric		
Precision (class 1)	0.4200	0.4200
Recall (class 1)	0.9200	0.9200
F1-Score (class 1)	0.5800	0.5800
AUC	0.9387	0.9393

Both models excelled at identifying positive cases. Because the computationally expensive RBF kernel offered no performance improvement over the much faster and simpler linear kernel, the **Linear SVM** is the recommended model from the SVM analysis.

Comparative Analysis: SVM vs. Balanced Decision Tree

The SVM results were compared against an updated Decision Tree analysis which also integrated the `class_weight='balanced'` parameter to create a fair comparison. The previous Decision Tree Analysis was recreated taking into account the class imbalance – which was not included in the original analysis. This allows for a more direct comparison between the SVM & DT models. The Decision Tree models were further optimized by pruning to manage the bias-variance tradeoff.

DT vs SVM Comparison

When placing the best-performing model from each analysis side-by-side, the results are statistically similar with one key difference:

Metric	Linear SVM (Balanced)	Decision Tree (Balanced, min_samples_leaf=35)
Recall	0.92	0.92
Precision	0.42	0.43
F1-Score	0.58	0.58
AUC	0.939	0.881
Overall Accuracy	0.85	0.85

The models appear to perform almost identically on the primary metrics: Recall (0.92), F1-Score (0.58), and Overall Accuracy (0.85) are all matched.

However, the Area Under the Curve (AUC), which measures a model's total discriminative power across all possible classification thresholds, provides the most significant finding. The Linear SVM (AUC = 0.939) is a fundamentally more robust classifier than the Decision Tree (AUC = 0.881). This substantial difference indicates that the SVM is superior at correctly distinguishing between the 'yes' and 'no' classes. While the Decision Tree matches the SVM's F1-score at the default threshold, its underlying ability to separate the classes is significantly weaker.

Final Recommendation

Based on this direct, balanced comparison, the **Linear SVM** is the recommended model.

While a superficial comparison metrics like the F1-Score (0.58) and Recall (0.92) suggests the models offer interchangeable performance, the Area Under the Curve (AUC) metric, which measures a model's total ability to discriminate between positive and negative classes across all thresholds, is the key differentiator.

The Linear SVM achieves an AUC of 0.939 and demonstrates a highly robust and reliable capacity to separate potential subscribers from non-subscribers. In contrast, the Decision Tree's respectable AUC of 0.881 is substantially lower. This discrepancy indicates that the Linear SVM is a more effective classification model.

Although the Decision Tree offers the significant practical advantage of interpretability—a key consideration for a financial institution that must explain decisions to regulators and stakeholders—this benefit does not outweigh the SVM's demonstrably superior predictive power – *for this use case*. If the core objective of the bank is to build the most accurate and reliable classifier, the Linear SVM is the overall recommended model.

Review of SVM and Decision Tree Models for COVID 19 Detection

In “*Decision Tree Ensembles to Predict Coronavirus Disease 2019 Infection*”, the authors focus on decision-tree-based ensembles (Random Forest, Bagging, AdaBoost, XGBoost, and imbalance-oriented models like Balanced Random Forest, RUSBagging, SMOTEBoost, RUSBoost). The key insights were that ensembles explicitly designed for imbalanced data outperform standard tree ensembles on recall, F1, AUROC, and especially AUPRC, making them better suited than generic models for this medical setting. Additionally, the choice of metric matters: AUPRC is more discriminating than AUROC on this imbalanced dataset and changes which tree ensemble appears best, suggesting that model selection can be misleading if you rely on the wrong metric.

In “*A novel approach to predict COVID-19 using support vector machine*”, the authors build a symptom-based dataset and train a linear SVM to classify patients into three classes: not infected, mildly infected, and severely infected. The key insights were that the SVM model achieves high accuracy overall, with very high precision, recall, and F1 for the “severely infected” class; researchers prioritized the severe class to minimize missed critical cases. In a comparative study, SVM outperformed other supervised models, including decision trees, Random Forest, AdaBoost, Naive Bayes, and kNN, on metrics like ROC area, accuracy, precision, recall, and F1; the authors recommend SVM as the preferred classifier for this symptom-based COVID-19 severity prediction task. They argue that SVM’s ability to handle nonlinearly separable patterns via the kernel trick and its robustness with relatively small, noisy clinical datasets make it more reliable than tree-based models given this set of conditions and data.

Review of SVM and Decision Trees in Project Management (Professional Area of Interest)

Project management uses predictive analytics to manage high-stakes risks like cost overruns, moving beyond purely qualitative expert judgment – which is common in the discipline and typical for practitioners. Machine learning models like Support Vector Machines and Decision Trees are useful as they can classify complex outcomes such as project failure, risk category, etc using the mixed categorical data typical of projects and project managed efforts . SVMs can handle complex, non-linear data; DTs and their ensembles (like Random Forests) provide interpretability. This interpretability is vital for Project Management Offices (PMO) who need to justify project decisions and mitigation strategies.

Prediction of Failures in the Project Management Knowledge Areas Using Optimized Ensemble Models

<https://link.springer.com/article/10.1007/s42452-025-07337-y>

The article evaluates classical ML models and optimized ensemble techniques to predict failures across industry standard PM knowledge areas using real project data. SVM (93.6%) and Decision Trees (93.6%) perform best among classical models, but ensemble methods achieve the highest accuracy (94%). Feature importance reveals that “Reason for Failure” and project manager education dominate predictive power; HR-related failures are the hardest to classify. The study shows that predictive

modeling can highlight failure-prone areas early and that interpretable or ensemble tree models offer stability and explanatory value for PMOs managing complex project portfolios. [OBJ]

Machine Learning in Project Analytics: A Data-Driven Framework and Case Study

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9463441/>

This study presents a ML framework for project analytics and uses it to predict construction project cost-overrun frequency. Random Forest initially performs best (77.5% accuracy), and feature-selection techniques increase accuracy to 85%; this highlights how important variable reduction is in project data. Delay-related and financial governance factors emerge as the strongest predictors for construction cost overruns. The authors argue that ML augments traditional tools like Earned Value Management (EVM) measurements by capturing nonlinear patterns and enabling earlier forecasting of schedule and cost stress.

Agile Project Status Prediction Using Interpretable Machine Learning

https://pure.port.ac.uk/ws/portalfiles/portal/92335910/Agile_project_status_prediction_using_interpretable_machine_learning.pdf

This study develops an interpretable Decision Tree model to predict agile IT project outcomes (“failed,” “delayed,” “success”) using 28 agility-based features from 360 telecom projects. Team capability, sprint-goal achievement, velocity, team morale, and experience emerge as the strongest predictors of project success. The Decision Tree achieves 97% accuracy, significantly outperforming SVM (71%) and KNN (82%)’ the interpretability of DT models is a key business benefit offering both superior accuracy and transparent rules for project assessment. The authors emphasize that early identification of failing agile projects enables resource reallocation and operational efficiency; they position interpretable machine learning as essential for real-world project governance. [OBJ]